

Artificial Intelligence

CSIT — 4th Semester — 2082 — Answer Sheet
Subject Code: CSC266

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What are the problems of depth limited search? Use the Alpha-Beta pruning algorithm to prune the following game tree assuming child nodes are visited from left to right. Show all final alpha and beta values computed at root, each internal node explored, and at the top of pruned branches.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How do you infer knowledge from semantic net? Illustrate with an example. Consider the following Bayesian network with random variables A, B, C and D. If A is true, find the probability of D being true.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is model free reinforcement learning? Differentiate between active and passive reinforcement learning. Describe the different operators used in genetic algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. How do you define rational? Can AI choose between right and wrong? Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What do you mean by PEAS description? Give an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Apply the hill climbing search algorithm to solve the following problem. Assume the heuristic function $h(n) = +1$ for all blocks in the support structure if the block is correctly positioned otherwise - 1.[5]Initial State ? A D C B Goal State ? D C B A

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Why do we need posterior probability? Design a semantic network for following statements. Dogs hate cats. Tom is a cat. Puppy is a dog. Cats chase rats. Rats are clever.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Design a Hebb net to implement logical OR function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Given the premises, all living things are either animal or plant. Plant needs sunlight for photosynthesis. Mustard is a living thing but it is not an animal. Does mustard need sunlight for photosynthesis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What are the phases for developing expert system? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Why do machines need to understand and generate natural language? What might be the challenges? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. List any one example of static and dynamic environment for agent. Why and when do you prefer rule based systems?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.